KERALA DIPLOMA · REVISION 2026

Practicum | Semester I | 4.5 Credits | 90 Contact Hours

Fundamentals of Food Processing Technology

Course 1421 — Introductory core subject covering macro & micronutrients, water activity, carbohydrates, proteins, lipids, vitamins, minerals, and practical food processing operations.

PROGRAMME	Food Processing Technology
COURSE CODE	1421
SEMESTER	I
TEACHING SCHEME	3:0:3:0 (L:T:P:S)
CREDITS	4.5
CIE MARKS	40
ESE MARKS	100

Official Source Declaration

This handbook is based on the official Kerala Polytechnic Revision 2026 syllabus for Course 1421 — Fundamentals of Food Processing Technology.

Verified inconsistencies disclosed: The syllabus document contains minor inconsistencies in module numbering (Module I appears twice in the major topics table) and practical hour allocation (listed as 0P in the module overview but includes practical experiments in the detailed syllabus). This handbook treats the detailed syllabus as authoritative and distributes practical content across all four modules as specified in the subtopic listings.

All module topics, subtopics, learning outcomes, practical experiments, self-learning activities, and assessment rules are extracted directly from the PDF. Educational elaboration (explanations, examples, diagrams, worked problems, calculators) is provided for teaching purposes and is not sourced from the syllabus.

Course Dashboard

At-a-glance view of course structure, teaching scheme, and assessment.

📘 Introduction

Introductory core subject for first-semester Diploma students. Provides basic knowledge of food components and their role in food processing and preservation.

🎯 Course Type

Practicum — integrates theory with practical sessions, laboratory experiments, food product preparation, and analytical techniques.

🔗 Pre-requisites

Basic chemistry concepts (acids, bases, solutions); awareness of common food materials, nutrients, and laboratory safety practices.

Teaching and Assessment Scheme

L	T	P	S	Self-learning (hrs/wk)	Total Hrs	Credits	CIE (Theory)	ESE (Theory)	CIE (Practical)	ESE (Practical)	Total
3	0	3	0	6	90	4.5	20	60	20	40	140

Legend: L — Lecture (1 credit per hour), T — Tutorial (1 credit), P — Practical (0.5 credit), S — Project (0.5 credit). CIE — Continuous Internal Evaluation, ESE — End Semester Examination.

How to Use This Handbook

1

Understand

Read each module's theory topics with deep explanations, diagrams, and Malayalam support notes.

2

Visualise

Study the SVG diagrams, formula cards, and comparison tables to reinforce concepts.

3

Apply

Work through the practical experiments, calculators, and worked examples for each topic.

4

Revise

Use the module-wise Q&A, revision cards, common mistakes list, and model paper for exam preparation.

Course Outcomes & CO–PO Mapping

What you will be able to do after completing this course.

Course Outcomes (CO) with Bloom's Taxonomy Level

CO	Course Outcome	Bloom's Level
CO1	Explain the composition, classification, and functional role of major food constituents.	Understand
CO2	Describe the properties of water and its significance in food processing and preservation.	Understand
CO3	Perform basic food processing operations and preparation of selected food products.	Apply
CO4	Apply simple analytical and sensory evaluation techniques to assess food quality.	Apply

CO–PO Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	1	-	-	-
CO2	3	2	-	-	1	-	-	-	-	-	1
CO3	2	2	3	2	2	1	-	1	-	-	-
CO4	2	3	-	3	2	-	-	-	1	-	1

Legends: 1 — Low; 2 — Medium; 3 — High; No Correlation — “-”

Curriculum Coverage Map

Every module, topic, subtopic, and practical mapped to COs and instructional hours.

Module-wise Coverage

Module	Title	Topics	L Hrs	P Hrs	CO
I	Fundamentals of Water and Nutrients in Food Systems	Macro/micronutrients; water activity; forms/structure/properties of water; food lab safety; moisture content; water activity; blanching effect	10	0	CO1
II	Chemistry, Properties, and Functional Role of Carbohydrates	Chemistry of carbohydrates; composition/structure; classification/properties; sources/functional importance; qualitative tests; TSS; jam preparation	12	0	CO2
III	Structure, Classification, and Functional Role of Proteins in Food	Functional role of proteins; protein structure; protein classification; mayonnaise; protein detection; protein estimation; paneer preparation	12	0	CO3
IV	Lipids, Vitamins, and Minerals: Properties and Functional Roles in Food	Lipids classification/properties; vitamins types/sources; minerals types/role; fat/oil detection; toffee; tomato ketchup; sauerkraut	9	0	CO4

Module Connection Roadmap

How the four modules build on each other.

I

Water & Nutrients

Foundation: food components, water activity, moisture analysis

II

Carbohydrates

Chemistry, classification, properties, jam preparation, TSS

III

Proteins

Structure, functional role, emulsion, coagulation, paneer

IV

Lipids, Vitamins, Minerals

Fats, oils, vitamins, minerals, fermented products, ketchup

MODULE I

Fundamentals of Water and Nutrients in Food Systems

L: 10 **P:** 0 **CO:** CO1 **Bloom:** Understand

This module introduces the fundamental components of food — macro and micronutrients — and the critical role of water in food systems. You will learn about water activity, its influence on food quality and shelf life, and the various forms and properties of water in food.

Macro and Micro Nutrients in Food

Definition and Importance

Food provides the body with essential nutrients that are classified into two broad categories based on the quantity required by the human body: **macronutrients** and **micronutrients**.

- **Macronutrients** — required in large amounts (grams per day). They provide energy and are the building blocks of body tissues. Includes **carbohydrates**, **proteins**, and **lipids (fats)**.
- **Micronutrients** — required in small amounts (milligrams or micrograms per day). They are essential for metabolic processes, enzyme function, and immune response. Includes **vitamins** and **minerals**.

Macronutrient Overview

Macronutrients at a Glance

Nutrient	Primary Role	Energy (kcal/g)	Food Sources
Carbohydrates	Primary energy source	4	Cereals, fruits, vegetables, sugar
Proteins	Growth, repair, enzymes	4	Meat, pulses, milk, eggs
Lipids	Energy storage, cell membranes	9	Oils, butter, nuts, fatty fish

Micronutrient Overview

- **Vitamins** — organic compounds required in trace amounts. Classified as fat-soluble (A, D, E, K) and water-soluble (B-complex, C). Essential for vision, immunity, bone health, and energy metabolism.
- **Minerals** — inorganic elements required for various bodily functions. Macro-minerals (Ca, P, Mg, Na, K, Cl) and trace minerals (Fe, Zn, Cu, I, Se, F).

Malayalam support: [illegible]

Tip: Remember the energy values — carbs and proteins provide 4 kcal/g, while fats provide 9 kcal/g. This is why high-fat foods are energy-dense.

Water Activity (a_w) and Its Importance

What is Water Activity?

Water activity (a_w) is the ratio of the partial pressure of water in a food to the partial pressure of pure water at the same temperature. It is a measure of the **free water** available in a food system for microbial growth, chemical reactions, and enzymatic activity.

$$a_w = P / P_0$$

where P = partial pressure of water in the food; P_0 = partial pressure of pure water at the same temperature.

Pure water has $a_w = 1.0$. Most fresh foods have $a_w > 0.95$, while dried foods have $a_w < 0.6$.

Significance in Food Processing and Preservation

- **Microbial growth:** Most bacteria require $a_w > 0.90$; yeasts and moulds can grow at a_w as low as 0.80 and 0.70 respectively. Lowering a_w inhibits spoilage.
- **Chemical reactions:** Enzymatic browning, Maillard reaction, lipid oxidation, and vitamin degradation are influenced by a_w.
- **Texture and mouthfeel:** Water activity affects crispness, softness, and chewiness of foods.
- **Shelf life:** Foods with low a_w (e.g., biscuits, cereals) have longer shelf life.

Water Activity and Food Stability

a_w Range	Microbial Growth	Examples
> 0.95	Most bacteria grow rapidly	Fresh meat, milk, vegetables
0.91–0.95	Salmonella, Staphylococcus aureus	Cheese, cured meat
0.80–0.91	Yeasts, moulds	Jams, syrups
0.60–0.80	Osmophilic yeasts, xerophilic moulds	Dried fruit, honey
< 0.60	No microbial growth	Biscuits, cereals, spices

Malayalam support: Water activity (a_w) [illegible]. [illegible]. a_w [illegible].

Forms, Structure, and Properties of Water in Food

Molecular Structure of Water

Water (H_2O) is a polar molecule with a bent structure. The oxygen atom is more electronegative, creating a partial negative charge near the oxygen and partial positive charges near the hydrogen atoms. This polarity allows water to form **hydrogen bonds**, giving it unique properties.

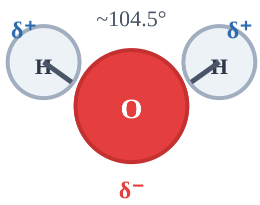

Fig. 1.1 — Water molecule showing bent structure, partial charges (δ^+, δ^-), and bond angle ~104.5°.

Forms of Water in Food

- **Free water:** Unbound water that is easily removed and available for microbial growth. Present in intercellular spaces and as a solvent.
- **Bound water:** Water that is physically or chemically associated with food components (proteins, carbohydrates, salts). Not easily removed and unavailable for microbial growth.
- **Hydration water:** Water bound to macromolecules through hydrogen bonds.
- **Capillary water:** Water held in capillaries and pores of food materials.

Key Properties of Water in Food Systems

- **Heat capacity:** High specific heat (4.18 J/g°C) — water resists temperature changes, important for food processing (heating, cooling).
- **Latent heat:** High latent heat of vaporisation (2257 kJ/kg at 100°C) — evaporation requires significant energy, relevant for drying and concentration.
- **Solvent properties:** Water dissolves many food components (sugars, salts, proteins) and acts as a medium for chemical reactions.
- **Reactivity:** Water participates in hydrolysis reactions, affecting starch, proteins, and lipids.

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Practical Experiments — Module I

•Exp 1.1 — Identification and Classification of Food Samples

Objective: Classify common food samples based on their major macro and micronutrient composition.

Procedure: Observe the given food samples (e.g., rice, milk, oil, fruits) and classify them based on their dominant nutrient — carbohydrate-rich, protein-rich, or fat-rich. Record in a tabular form.

Sample Classification

Food Sample	Dominant Nutrient	Category
Rice	Carbohydrate	Macronutrient
Milk	Protein + Fat	Macronutrient
Groundnut oil	Fat	Macronutrient
Spinach	Vitamins, Minerals	Micronutrient

•Exp 1.2 — Qualitative Identification of Major Nutrients

Objective: Perform simple qualitative tests to identify starch, protein, and fat in food samples.

- **Starch test (Iodine test):** Add a few drops of iodine solution to the sample. Blue-black colour indicates starch.
- **Protein test (Biuret test):** Add 1 mL of 10% NaOH and 2–3 drops of 1% $CuSO_4$ to the sample. Violet colour indicates protein.
- **Fat test (Spot test):** Rub the sample on filter paper. A translucent spot indicates fat.

> **Tip:** Always use a fresh sample for each test and record observations immediately.

Exp 1.3 — Determination of Water Activity

Objective: Determine the water activity of selected food samples and relate it to food quality and shelf life.

Method: Use a water activity meter (e.g., Aw meter) or the dew-point method. Place the sample in the measuring chamber, allow equilibration, and record the a_w value.

Interpretation: Compare a_w values of fresh (a_w ~0.98) and dried (a_w ~0.4–0.6) samples. Relate to microbial stability.

•Exp 1.4 — Moisture Content by Oven Drying

Objective: Determine the moisture content of food samples using the oven drying method.

Procedure: Weigh a clean, dried moisture dish. Add ~5 g of the sample, weigh accurately. Place in an oven at 105°C for 3–4 hours. Cool in a desiccator and re-weigh. Repeat until constant weight.

$$\textbf{Moisture (\%)} = [(W_{initial} - W_{dry}) / W_{initial}] \times 100$$

$W_{initial}$ = weight of sample before drying; W_{dry} = weight of sample after drying.

Given: Sample weight = 5.2 g; Dry weight = 4.8 g.

Required: Moisture content (%)

Calculation: (5.2 − 4.8) / 5.2 × 100 = 0.4 / 5.2 × 100 = 7.69%

Answer: Moisture content = **7.69%**

🧪 **Live Moisture Calculator**

Initial weight (g) 5.2

Dry weight (g) 4.8

Moisture = 7.69%

Safety: Use oven gloves when handling hot dishes. Ensure the oven is well-ventilated. Do not overheat samples that may catch fire.

Module I Summary: Food contains macro (carbohydrates, proteins, fats) and micro (vitamins, minerals) nutrients. Water exists in free and bound forms, and water activity (a_w) is critical for microbial stability. Practical experiments include moisture determination by oven drying, a_w measurement, and qualitative nutrient tests.

MODULE II

Chemistry, Properties, and Functional Role of Carbohydrates

L: 12 **P:** 0 **CO:** CO2 **Bloom:** Understand

Carbohydrates are the most abundant organic compounds in nature and the primary energy source in the human diet. This module covers their chemistry, classification, properties, functional roles in food processing, and practical applications including jam and toffee preparation.

Chemistry of Carbohydrates

Definition and General Formula

Carbohydrates are polyhydroxy aldehydes or ketones, or compounds that yield such derivatives on hydrolysis. The general formula is $\mathbf{C_n(H_2O)_n}$ (where $n \geq 3$).

- **Monosaccharides:** Simple sugars, cannot be hydrolysed further. Examples: glucose ($C_6H_{12}O_6$), fructose, galactose.
- **Oligosaccharides:** 2–10 monosaccharide units linked by glycosidic bonds. Examples: sucrose (glucose + fructose), maltose (glucose + glucose).
- **Polysaccharides:** More than 10 monosaccharide units. Examples: starch, cellulose, glycogen.

Functional Groups

- **Aldehyde group (−CHO):** Present in aldoses (e.g., glucose).
- **Ketone group (C=O):** Present in ketoses (e.g., fructose).
- **Hydroxyl groups (−OH):** Multiple hydroxyl groups make carbohydrates polar and water-soluble.

Linear form:

CHO
|
H—C—OH
|
HO—C—H ...

Ring form (α-D-glucopyranose):

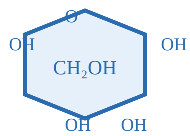

Fig. 2.1 — Glucose: linear (left) and ring (pyranose) forms.

Malayalam support: [illegible]

Composition and Structure of Carbohydrates

Composition

Carbohydrates are composed of **carbon (C)**, **hydrogen (H)**, and **oxygen (O)** in the ratio $(CH_2O)_n$. They may also contain nitrogen (in amino sugars) or sulfur (in some polysaccharides).

Structural Forms

- **Straight-chain (acyclic) form:** Open chain with aldehyde or ketone group.
- **Ring (cyclic) form:** The majority of carbohydrates exist in ring form in aqueous solution — pyranose (6-membered) or furanose (5-membered) rings.
- **α and β anomers:** The orientation of the OH group at the anomeric carbon (C1) determines α or β configuration.

Common Monosaccharides and Their Structure

Name	Formula	Type	Ring Form
Glucose	$C_6H_{12}O_6$	Aldose (hexose)	Pyranose
Fructose	$C_6H_{12}O_6$	Ketose (hexose)	Furanose
Galactose	$C_6H_{12}O_6$	Aldose (hexose)	Pyranose

Classification and Properties of Carbohydrates

Classification Based on Hydrolysis

- **Monosaccharides:** Glucose, fructose, galactose, ribose. Sweet, water-soluble, reducing sugars.
- **Disaccharides:** Sucrose (glucose + fructose), maltose (glucose + glucose), lactose (glucose + galactose). Hydrolysed to monosaccharides.
- **Oligosaccharides:** Raffinose (galactose + glucose + fructose), stachyose.
- **Polysaccharides:** Starch (amylose + amylopectin), glycogen, cellulose, pectin. Non-sweet, mostly insoluble, serve as structural or storage molecules.

Key Properties

- **Solubility:** Small carbohydrates are soluble in water due to hydroxyl groups; high molecular weight polysaccharides are less soluble.
- **Sweetness:** Fructose > sucrose > glucose > maltose > lactose.
- **Reducing property:** Carbohydrates with free aldehyde or ketone group can reduce Fehling's or Benedict's solution.
- **Hydrolysis:** Disaccharides and polysaccharides are hydrolysed by acids or enzymes to monosaccharides.
- **Gelation:** Some polysaccharides (e.g., pectin, starch) form gels, used in jams, jellies, and sauces.

Properties of Major Carbohydrate Classes

Class	Examples	Sweetness	Reducing	Solubility
Monosaccharides	Glucose, Fructose	High	Yes	High
Disaccharides	Sucrose, Maltose	Moderate	Variable	High
Polysaccharides	Starch, Cellulose	None	No	Low

Sources and Functional Importance of Carbohydrates

Food Sources

- **Cereals and grains:** Rice, wheat, maize, oats (starch).
- **Fruits:** Fructose, glucose, sucrose (natural sugars).
- **Vegetables:** Potatoes, sweet potatoes, carrots (starch, sugars).
- **Legumes:** Beans, lentils (starch, oligosaccharides).
- **Dairy:** Lactose (milk sugar).

Functional Role in Food Processing

- **Sweetening:** Sugars provide sweetness, flavour enhancement.
- **Texture and mouthfeel:** Starch and pectin contribute to viscosity, gelation, and body.
- **Browning:** Maillard reaction (sugars + amino acids) and caramelisation (sugars alone) produce colour and flavour.
- **Preservation:** High sugar concentrations reduce water activity, inhibiting microbial growth (e.g., jams, jellies).
- **Fermentation:** Sugars are substrates for fermentation (alcohol, bread, yogurt).
- **Emulsification and stabilisation:** Some carbohydrates act as thickeners and stabilisers.

> **Malayalam support:** [illegible]

Practical Experiments — Module II

•Exp 2.1 — Molisch's and Benedict's Tests

Objective: Perform qualitative tests to identify carbohydrates in food samples.

- **Molisch's test:** Add 2–3 drops of Molisch's reagent (α-naphthol in ethanol) to the sample, then carefully add concentrated H_2SO_4 along the side of the test tube. A purple-violet ring at the interface indicates carbohydrate.
- **Benedict's test:** Add 2 mL of Benedict's reagent to the sample and heat in a boiling water bath for 5 minutes. A green, yellow, or brick-red precipitate indicates reducing sugar.

> **Note:** Molisch's test is a general test for all carbohydrates; Benedict's test distinguishes reducing sugars from non-reducing sugars.

•Exp 2.2 — Iodine Test for Starch

Objective: Identify the presence of starch in different food samples using iodine test.

Procedure: Add 2–3 drops of dilute iodine solution (iodine in potassium iodide) to the sample. A **blue-black colour** indicates the presence of starch (amylose forms a complex with iodine).

Examples: Potato, rice, wheat flour give positive (blue-black) results; sugar, oil, salt give negative (no colour change).

📊 TSS to Sugar Content Estimator

°Brix (TSS): 65

Sample volume (mL): 100

Sugar content ≈ 65.0 g per 100 mL

•Exp 2.3 — Determination of Total Soluble Solids (TSS)

Objective: Determine the total soluble solids (TSS) of sugar solutions using a hand refractometer.

Procedure: Place a drop of the sample on the refractometer prism, close the cover, and read the TSS value (in °Brix) through the eyepiece. Clean the prism with distilled water after each reading.

Interpretation: TSS is a measure of the sugar content in solutions. For jams, TSS should be around 65–68 °Brix for proper gelation and preservation.

> **Tip:** Calibrate the refractometer with distilled water (0 °Brix) before use. Ensure the sample is at room temperature for accurate readings.

•Exp 2.4 — Preparation of Jam and Evaluation of Quality

Objective: Prepare jam using fruit pulp and sugar, and relate the functional role of carbohydrates in product quality.

Procedure:

- Take 1 kg of fruit pulp (e.g., mango, pineapple) with 0.75–1 kg of sugar (depending on fruit acidity).
- Add 5–8 g of pectin (if needed) and 3–5 mL of citric acid solution (to adjust pH to ~3.2–3.4).
- Cook with constant stirring until the TSS reaches 65–68 °Brix and the "sheet test" is positive (a drop of syrup forms a sheet when cooled).
- Pour into sterilised jars, seal, and cool.

Quality evaluation: Assess colour, texture, spreadability, flavour, and TSS. Relate to carbohydrate (sugar, pectin) functionality in gelation and preservation.

> **Safety:** Use gloves and eye protection when handling hot syrup. Avoid splashing of hot sugar solution.

> **Module II Summary:** Carbohydrates are polyhydroxy aldehydes/ketones classified as mono-, di-, oligo-, and polysaccharides. They provide sweetness, texture, browning, and preservation. Practicals include qualitative tests, TSS measurement, and jam preparation where sugar and pectin form a gel.

MODULE III

Structure, Classification, and Functional Role of Proteins in Food

L: 12 **P:** 0 **CO:** CO3 **Bloom:** Apply

Proteins are complex macromolecules made of amino acids. This module covers the structure of proteins, their classification, functional roles in food processing, and practical applications including mayonnaise, paneer, and protein estimation.

•Functional Role of Proteins in Food Processing

Proteins perform numerous functions in food systems due to their amphiphilic nature and ability to form gels, foams, and emulsions.

- **Emulsification:** Proteins reduce interfacial tension between oil and water, stabilising emulsions (e.g., mayonnaise, salad dressings).
- **Gelation:** Proteins form three-dimensional networks that trap water, providing structure and texture (e.g., paneer, tofu, cooked meat).
- **Foaming:** Proteins form stable foams by forming films around air bubbles (e.g., meringue, whipped cream).
- **Water holding capacity:** Proteins bind water, improving juiciness and texture of meat, fish, and bakery products.
- **Coagulation:** Denatured proteins aggregate to form solid structures (e.g., cheese, yogurt).
- **Nutritional value:** Proteins provide essential amino acids required for growth and maintenance.
- **Flavour binding:** Proteins bind flavour compounds, influencing taste and aroma.

> **Malayalam support:** □□□□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□□□ (mayonnaise □□□□□□□□□), □□□□□□□ (paneer, tofu), □□□□□□□□ (meringue), □□□□□□□ (meat, bakery) □□□□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□□□□. □□□□□□□ □□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□ □□□, □□□□□, □□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□.

·Structure of Proteins

Four Levels of Protein Structure

- **Primary structure:** The linear sequence of amino acids joined by peptide bonds. Determined by the gene sequence.
- **Secondary structure:** Folding of the polypeptide chain into α-helices or β-pleated sheets, stabilised by hydrogen bonds.
- **Tertiary structure:** Three-dimensional folding of a single polypeptide chain, stabilised by hydrophobic interactions, disulfide bonds, and ionic bonds.
- **Quaternary structure:** Association of two or more polypeptide chains (subunits) to form a functional protein (e.g., haemoglobin).

Fig. 3.1 — Four levels of protein structure: primary (linear), secondary (helix), tertiary (folded), quaternary (multi-chain assembly).

> **Note:** Denaturation disrupts secondary, tertiary, and quaternary structures without breaking peptide bonds (primary structure).

Classification of Proteins

Based on Composition

- **Simple proteins:** Composed only of amino acids. Examples: albumins, globulins, glutelins, prolamins.
- **Conjugated proteins:** Contain non-protein groups (prosthetic groups). Examples: glycoproteins (carbohydrate), lipoproteins (lipid), metalloproteins (metal ions), nucleoproteins (nucleic acid).
- **Derived proteins:** Produced by chemical or enzymatic hydrolysis of simple/conjugated proteins. Examples: peptones, peptides, proteoses.

Based on Shape

- **Globular proteins:** Compact, spherical shape, soluble in water. Examples: egg albumin, serum albumin, enzymes.
- **Fibrous proteins:** Long, thread-like, insoluble in water. Provide structural support. Examples: collagen, keratin, myosin.

Classification of Food Proteins

Type	Examples	Solubility	Function
Albumins	Egg white, serum albumin	Water soluble	Nutritional, foaming
Globulins	Legume proteins, myosin	Salt soluble	Emulsification, gelation
Glutelins	Wheat gluten (glutenin)	Acid/alkali soluble	Dough elasticity
Prolamins	Gliadin (wheat), zein (maize)	Alcohol soluble	Viscosity, film formation

Practical Experiments — Module III

•Exp 3.1 — Preparation of Mayonnaise and Study of Emulsion Stability

Objective: Prepare mayonnaise and relate the emulsifying property of proteins to product stability.

Procedure:

- Take egg yolk (contains lipoproteins and lecithin) in a bowl.
- Add mustard powder, salt, sugar, and vinegar (or lemon juice).
- Slowly add oil (e.g., sunflower oil) in a thin stream while continuously whisking.
- Continue until a thick, stable emulsion is formed.
- Observe the stability over time (e.g., no oil separation).

Role of proteins: Egg yolk lipoproteins adsorb at the oil-water interface, forming a protective film that prevents coalescence (emulsion stability).

> **Warning:** Use pasteurised eggs to reduce the risk of Salmonella. Store mayonnaise at refrigeration temperature (≤4°C).

•Exp 3.2 — Detection of Proteins Using Qualitative Tests

Objective: Identify proteins in food samples using standard qualitative tests.

- **Biuret test:** Add 1 mL of 10% NaOH and 2–3 drops of 1% $CuSO_4$ to the sample. Violet or purple colour indicates protein.
- **Xanthoproteic test:** Add concentrated HNO_3 to the sample; a yellow colour develops which turns orange on adding alkali. Indicates aromatic amino acids (tyrosine, tryptophan).
- **Millon's test:** Add Millon's reagent ($HgNO_3$ in HNO_3) to the sample; white precipitate turns red on heating. Indicates tyrosine.
- **Ninhydrin test:** Add ninhydrin reagent and heat. Blue-purple colour indicates amino acids/proteins.

> **Safety:** Handle HNO_3 and Hg-containing reagents in a fume hood with gloves and eye protection.

Exp 3.3 — Estimation of Protein Content (Kjeldahl Method)

Objective: Estimate the protein content in a given food sample using the Kjeldahl method.

Principle: The protein is digested with concentrated H_2SO_4 in the presence of a catalyst, converting organic nitrogen to ammonium sulfate. The ammonia is distilled and titrated to determine the nitrogen content, which is then multiplied by a factor (usually 6.25) to obtain protein content.

Protein (%) = Nitrogen (%) × 6.25

Nitrogen content determined by titration; factor 6.25 is the average conversion factor for most proteins.

Given: Nitrogen content = 1.6%

Required: Protein content (%)

Calculation: 1.6 × 6.25 = 10.0%

Answer: Protein content = **10.0%**

🧪 **Live Protein Estimation Calculator**

Nitrogen (%)	Conversion factor
1.6	6.25

Protein = 10.00%

Exp 3.4 — Preparation of Paneer by Acid Coagulation

Objective: Prepare paneer from milk by acid coagulation and relate protein denaturation and coagulation to product formation.

Procedure:

- Heat 1 L of milk to 80–85°C (near boiling).
- Add 1–2% acid (citric acid or lemon juice) slowly with gentle stirring.
- Milk proteins (casein) coagulate and form curds.
- Allow to stand for 5–10 minutes.
- Strain through a muslin cloth, press to remove excess whey.
- Cool and cut into pieces.

Principle: Acid reduces the pH to the isoelectric point of casein (~pH 4.6), causing denaturation and aggregation of casein micelles to form a gel (coagulum).

> **Quality check:** Good paneer should be firm, white, and have a mild acidic taste. Yield depends on milk fat and protein content.

> **Module III Summary:** Proteins have four levels of structure and are classified by composition (simple/conjugated) and shape (globular/fibrous). They function in emulsification, gelation, foaming, and water holding. Practicals include mayonnaise (emulsion), paneer (coagulation), and protein detection/estimation.

MODULE IV

Lipids, Vitamins, and Minerals: Properties and Functional Roles in Food

L: 9 **P:** 0 **CO:** CO4 **Bloom:** Apply

Lipids (fats and oils) are energy-dense compounds with critical roles in texture, flavour, and nutrition. Vitamins and minerals are essential micronutrients. This module covers their classification, sources, functional roles, and practical applications including toffee, tomato ketchup, and sauerkraut.

Classification and Properties of Lipids

Definition

Lipids are a diverse group of organic compounds that are **insoluble in water** but soluble in non-polar organic solvents (e.g., ether, chloroform, acetone). They are esters of fatty acids and alcohols.

Classification

- **Simple lipids:** Esters of fatty acids with alcohols. Includes **fats (triglycerides)** — solid at room temperature, and **oils** — liquid at room temperature. Also includes waxes.
- **Compound lipids:** Contain additional groups besides fatty acids and alcohol. Includes **phospholipids** (e.g., lecithin), **glycolipids**, and **lipoproteins**.
- **Derived lipids:** Products of hydrolysis of simple/compound lipids. Includes **fatty acids**, **steroids** (cholesterol, vitamin D), and **carotenoids**.

Key Properties

- **Solubility:** Insoluble in water, soluble in organic solvents.
- **Melting point:** Fats (saturated) have higher melting points than oils (unsaturated).
- **Oxidation:** Unsaturated fatty acids are prone to oxidative rancidity, producing off-flavours.
- **Emulsification:** Lipids can be dispersed in water with the help of emulsifiers (e.g., phospholipids, proteins).
- **Energy density:** 9 kcal/g — the highest among macronutrients.

Comparison of Fats and Oils

Property	Fats	Oils
State at RT	Solid	Liquid
Fatty acid composition	Predominantly saturated	Predominantly unsaturated
Examples	Butter, ghee, lard	Sunflower oil, olive oil, fish oil
Melting point	Higher	Lower

> **Malayalam support:** [illegible] ([illegible]) [illegible] ([illegible]) [illegible]. 9 kcal/g [illegible], [illegible], [illegible].

Vitamins — Types and Sources

Definition

Vitamins are organic compounds required in small amounts for normal physiological functions. They are not synthesized in adequate amounts by the body and must be obtained through diet.

Classification

- **Fat-soluble vitamins:** A, D, E, K. Absorbed with dietary fats, stored in the liver and adipose tissue. Excess intake can be toxic.
- **Water-soluble vitamins:** B-complex (B_1, B_2, B_3, B_5, B_6, B_7, B_9, B_{12}) and vitamin C. Not stored in large amounts; excess excreted in urine.

Key Vitamins: Sources and Functions

Vitamin	Type	Dietary Sources	Primary Function
A	Fat-soluble	Carrots, liver, dairy	Vision, immunity, skin health
D	Fat-soluble	Sunlight, fish, fortified milk	Calcium absorption, bone health
E	Fat-soluble	Nuts, seeds, vegetable oils	Antioxidant, cell protection
K	Fat-soluble	Leafy greens, vegetable oils	Blood clotting
B_1 (Thiamine)	Water-soluble	Cereals, pulses, meat	Energy metabolism
B_2 (Riboflavin)	Water-soluble	Milk, eggs, green vegetables	Energy metabolism, vision
B_3 (Niacin)	Water-soluble	Meat, fish, cereals	Energy metabolism
B_6 (Pyridoxine)	Water-soluble	Meat, fish, bananas	Amino acid metabolism
B_{12} (Cobalamin)	Water-soluble	Animal products, fortified foods	Red blood cell formation, nerve function
C (Ascorbic acid)	Water-soluble	Citrus fruits, tomatoes, leafy greens	Antioxidant, collagen synthesis

> **Warning:** Fat-soluble vitamins (A, D, E, K) can accumulate in the body and cause toxicity if consumed in excess. Water-soluble vitamins are generally safe as excess is excreted.

Minerals — Types and Functional Roles

Definition

Minerals are inorganic elements required for various physiological functions. They are classified as **macro-minerals** (required >100 mg/day) and **micro-minerals (trace elements)** (required <100 mg/day).

Macro-Minerals: Sources and Functions

Mineral	Dietary Sources	Primary Function
Calcium (Ca)	Dairy, leafy greens, fortified foods	Bone/teeth formation, muscle contraction, nerve function
Phosphorus (P)	Dairy, meat, cereals	Bone/teeth formation, energy metabolism (ATP)
Magnesium (Mg)	Nuts, whole grains, leafy greens	Enzyme activation, muscle function, nerve function
Sodium (Na)	Table salt, processed foods	Fluid balance, nerve impulse transmission
Potassium (K)	Fruits, vegetables, legumes	Fluid balance, muscle contraction, nerve function

Trace Minerals: Sources and Functions

Mineral	Dietary Sources	Primary Function
Iron (Fe)	Meat, pulses, leafy greens	Oxygen transport (haemoglobin), energy metabolism
Zinc (Zn)	Meat, seafood, nuts	Immune function, wound healing, enzyme activity
Iodine (I)	Seafood, iodised salt	Thyroid hormone synthesis
Selenium (Se)	Fish, eggs, Brazil nuts	Antioxidant function (glutathione peroxidase)
Copper (Cu)	Seafood, nuts, liver	Iron metabolism, enzyme function

> **Malayalam support:** [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible]-[illegible] (Ca, P, Mg, Na, K) [illegible] [illegible]-[illegible] (Fe, Zn, I, Se, Cu) [illegible] [illegible] [illegible]. [illegible] [illegible], [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible].

Practical Experiments — Module IV

Exp 4.1 — Detection of Fats and Oils (Qualitative Tests)

Objective: Identify fats and oils in food samples using standard qualitative tests.

- **Spot test (Grease spot test):** Rub the sample on filter paper and hold it up to light. A translucent spot indicates the presence of fat/oil.
- **Emulsion test:** Shake the sample with water; a cloudy or milky appearance indicates fat/oil emulsified.
- **Acrolein test:** Heat a small amount of fat with potassium bisulfate ($KHSO_4$). A pungent, acrid smell of acrolein (from glycerol dehydration) confirms fat.

Exp 4.2 — Preparation of Toffee and Study of Role of Fat

Objective: Prepare toffee using sugar, milk solids, and fat, and study the role of fat in texture and mouthfeel.

Procedure:

- Mix sugar, milk solids, and fat (butter or margarine) in a pan with water.
- Heat with continuous stirring until the mixture reaches the "hard crack" stage (~150°C).
- Pour into a greased mould, cool, and cut into pieces.

Role of fat: Fat contributes to the **shortening effect** — it interferes with sugar crystallisation, giving a smooth, creamy texture and rich mouthfeel. Fat also enhances flavour release.

> **Safety:** Hot toffee mixture (>150°C) can cause severe burns. Use gloves and handle with care. Work in a well-ventilated area.

Exp 4.3 — Preparation of Tomato Ketchup

Objective: Prepare tomato ketchup with emphasis on ingredient composition.

Procedure:

- Take ripe tomatoes, wash, remove stems, and cut into pieces.
- Cook with a little water until soft, then strain to obtain tomato puree.
- Add sugar, salt, vinegar (or citric acid), and spices (like cinnamon, cloves, pepper).
- Cook with stirring until the desired consistency is reached (TSS ~30–32 °Brix).
- Fill into sterilised bottles and seal.

Key ingredients: Tomatoes (lycopene, vitamins, acids), sugar (sweetness, preservation), salt (flavour), vinegar (acidity, preservation).

Tip: Addition of a small amount of starch (e.g., 1–2%) can improve the consistency and prevent separation.

Exp 4.4 — Preparation of Sauerkraut by Natural Fermentation

Objective: Prepare sauerkraut by natural fermentation and understand the role of lactic acid bacteria.

Procedure:

- Shred fresh cabbage (white cabbage) finely.
- Mix with 2–2.5% salt (w/w) and pack tightly into a clean fermentation vessel.
- Cover with a plate or weight to keep the cabbage submerged in its own brine.
- Ferment at room temperature (18–22°C) for 2–4 weeks.
- Observe the development of sour flavour, colour change, and microbial activity.

Principle: Lactic acid bacteria (Leuconostoc, Lactobacillus) naturally present on cabbage convert sugars to lactic acid, lowering pH and preserving the product. Sauerkraut is rich in vitamin C, dietary fiber, and probiotics.

Safety: Ensure the cabbage is fully submerged to prevent mould growth. Use clean, sanitised equipment to avoid contamination.

Module IV Summary: Lipids (fats/oils) are energy-dense, insoluble compounds with functional roles in texture and flavour. Vitamins (fat- and water-soluble) and minerals (macro and trace) are essential micronutrients.

Practicals include fat detection, toffee (role of fat), tomato ketchup (ingredient composition), and sauerkraut (fermentation).

Formula Bank

Key formulas and equations from all modules, organised for quick reference.

$a_w = P / P_0$

Water activity: P = partial pressure in food, P_0 = pure water pressure

Moisture (%) = $[(W_i - W_d) / W_i] \times 100$

W_i = initial weight, W_d = dry weight

Carbohydrates: $C_n(H_2O)_n$

General formula for carbohydrates ($n \geq 3$)

Protein (%) = Nitrogen (%) × 6.25

Kjeldahl conversion factor for protein estimation

Energy from fats = 9 kcal/g

Energy density of lipids

Energy from carbs/proteins = 4 kcal/g

Energy density of carbohydrates and proteins

Diagram Library for Rapid Revision

Key diagrams from each module, linked by topic anchors.

🔬 Water Molecule

Fig. 1.1 — Water structure, partial charges, hydrogen bonding.
→ Module I

🧪 Glucose Structure

Fig. 2.1 — Linear and ring (pyranose) forms of glucose.
→ Module II

🧬 **Protein Structure**

Fig. 3.1 — Primary, secondary, tertiary, quaternary levels.
→ Module III

Self-Learning Activities

Official self-learning activities from the syllabus, mapped to modules.

📋 **Module I**

Study of food processing laboratory guidelines, safety measures, and basic equipment.

📋 **Module I**

Study the effect of blanching on selected vegetables (colour, texture, moisture loss).

📋 **Module II**

Detection of carbohydrates using Molisch's and Benedict's tests.

📋 **Module III**

Detection of proteins using standard qualitative tests (Biuret, Xanthoproteic).

📋 **Module IV**

Detection of fats and oils using qualitative tests (spot test, acrolein test).

📋 **Module IV**

Preparation of sauerkraut by natural fermentation and observation of microbial activity.

Assessment Methodology and Examination Pattern

Continuous Internal Evaluation (CIE) and End Semester Examination (ESE) structure.

Assessment Methodology — Practicum

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	ESE
Description	—	Average of Self-learning activities from each module	Continuous evaluation	Practical test (first half)	Practical test (second half)	Series Exam (2 modules)	Written Exam (theory) + Lab Exam
Duration	—	—	—	3 hours	3 hours	2 hours	2.5 hours + 3 hours
Exam mark	10	5	10	40	40	50	50+60
Converted to	10	10	10	10	10	10	—
Marks	10	10	10	10	10	10	40
Total	40 (CIE) + 100 (ESE) = 140						

Series Examination Pattern — Theory

Series Test Structure (2 hours)

Part	Marks	Type
Part A	21	Multiple/short answer (1 mark each)
Part B	33	Short answer (3 marks each)
Part C	27	Descriptive (7 marks each, with choice)
Total	**81**	—

End Semester Examination Pattern — Theory

ESE Pattern (2.5 hours)

Part	No. of Questions	Marks per Q	Total Marks
Part A (2 from each module)	8	1	8
Part B (2 out of 3 from each module)	8	3	24
Part C (1 set with choice from each module)	4	7	28
Total	—	—	**60**

CIE — Practical Continuous Evaluation: Criteria include preparation & punctuality (10), experimental setup & procedure (10), observation & data recording (5), analysis & result interpretation (10), viva voce (10), workmanship & discipline (5) — Total 50 marks.

Module-wise Questions with Answers

Practice questions organised by module, with detailed answers.

Module I — Water and Nutrients

Q: Define water activity and explain its significance in food preservation.

A:

Water activity (a_w) is the ratio of the partial pressure of water in a food to the partial pressure of pure water at the same temperature.

Significance:

- Determines microbial growth — most bacteria require $a_w > 0.90$; yeasts > 0.80; moulds > 0.70.
- Affects chemical reactions like browning, oxidation, and enzymatic activity.
- Lowering a_w (by drying, adding sugar/salt) extends shelf life.
- Foods with $a_w < 0.6$ are shelf-stable without refrigeration.

Q: Describe the forms and properties of water in food systems.

A:

Forms of water:

- **Free water:** Unbound, easily removed, available for microbial growth.
- **Bound water:** Associated with macromolecules, not easily removed, unavailable for microbes.

Properties: High specific heat (4.18 J/g°C), high latent heat of vaporisation (2257 kJ/kg at 100°C), excellent solvent, participates in hydrolysis reactions.

Q: What are macro and micronutrients? Give examples of each.

A:

Macronutrients — required in large amounts (g/day): Carbohydrates, proteins, lipids. Provide energy (4, 4, 9 kcal/g respectively).

Micronutrients — required in small amounts (mg/µg/day): Vitamins and minerals. Essential for metabolism, immunity, enzyme function.

Module II — Carbohydrates

Q: Classify carbohydrates with examples and describe their functional roles in food.

A:

Classification:

- **Monosaccharides:** Glucose, fructose, galactose.
- **Disaccharides:** Sucrose, maltose, lactose.
- **Oligosaccharides:** Raffinose, stachyose.
- **Polysaccharides:** Starch, cellulose, pectin, glycogen.

Functional roles: Sweetening, texture/gelation, browning (Maillard, caramelisation), preservation (water activity reduction), fermentation substrate.

Q: Explain the principle and procedure of jam preparation. What is the role of sugar and pectin?

A:

Procedure: Fruit pulp + sugar + pectin + acid → cook to 65–68 °Brix → pour into jars.

Role of sugar: Sweetness, lowers a_w for preservation, contributes to gel structure.

Role of pectin: Forms gel network with sugar and acid — high methoxyl pectin gels at low pH (~3.2) and high TSS (~65%).

Q: What is the significance of TSS measurement in food processing?

A:

TSS (Total Soluble Solids) is a measure of the dissolved solids (mainly sugars) in a solution, expressed in °Brix.

Significance: Indicates sweetness, determines the end-point in jam/jelly making, influences preservation (higher TSS lowers a_w), and affects consumer acceptance.

Module III — Proteins

Q: Describe the four levels of protein structure.

A:

- **Primary:** Linear sequence of amino acids linked by peptide bonds.
- **Secondary:** Folding into α-helices and β-sheets, stabilised by hydrogen bonds.
- **Tertiary:** 3D folding of a single polypeptide, stabilised by hydrophobic interactions, disulfide bonds, ionic bonds.
- **Quaternary:** Assembly of multiple polypeptide subunits (e.g., haemoglobin).

Q: Explain the functional roles of proteins in food processing with examples.

A:

- **Emulsification:** Egg yolk proteins in mayonnaise.
- **Gelation:** Milk casein in paneer, cheese.

- **Foaming:** Egg white proteins in meringue.
- **Water holding:** Myosin in meat, gluten in dough.
- **Coagulation:** Heat/acid denaturation in tofu, paneer.

Q: What is the principle of paneer preparation? How does acid cause coagulation?

A:

Principle: Acid (citric acid/lemon juice) lowers the pH of milk to the isoelectric point of casein (~pH 4.6). At this pH, casein micelles lose their charge, aggregate, and form a coagulum (curd).

Process: Heat milk to 80–85°C → add acid slowly → curd formation → strain → press → paneer.

Module IV — Lipids, Vitamins, Minerals

Q: Classify lipids and describe the properties of fats and oils.

A:

Classification:

- **Simple lipids:** Fats (solid), oils (liquid), waxes.
- **Compound lipids:** Phospholipids, glycolipids, lipoproteins.
- **Derived lipids:** Fatty acids, steroids, carotenoids.

Properties: Insoluble in water, soluble in organic solvents; high energy density (9 kcal/g); fats have higher melting points than oils; unsaturated lipids are prone to oxidative rancidity.

Q: Distinguish between fat-soluble and water-soluble vitamins. Give examples and sources.

A:

- **Fat-soluble (A, D, E, K):** Absorbed with fats, stored in liver/adipose, risk of toxicity. Sources: A (carrots, liver), D (sunlight, fish), E (nuts, oils), K (leafy greens).
- **Water-soluble (B-complex, C):** Not stored, excess excreted, low toxicity risk. Sources: B_1 (cereals), B_2 (milk), B_{12} (animal products), C (citrus, tomatoes).

Q: Explain the role of minerals in human nutrition with examples of macro and trace minerals.

A:

Macro-minerals (>100 mg/day): Calcium (bone), phosphorus (ATP), magnesium (enzyme), sodium/potassium (fluid balance).

Trace minerals (<100 mg/day): Iron (haemoglobin), zinc (immunity), iodine (thyroid), selenium (antioxidant), copper (iron metabolism).

Practice Model Paper

Based on the official examination pattern — not an official SITTTR model question paper. Use for practice.

End Semester Examination — Theory (2.5 hours)

Maximum Marks: 60

Part A — (8 × 1 = 8 marks)

Answer **all** questions (2 from each module). Each question carries 1 mark.

- 1. Define water activity.
- 2. Give two examples of macronutrients.
- 3. Write the general formula of carbohydrates.
- 4. What is the principle of Benedict's test?
- 5. What is the isoelectric point of casein?
- 6. Name two globular proteins.
- 7. Classify vitamins based on solubility.
- 8. Give two sources of iron in the diet.

Part B — (8 × 3 = 24 marks)

Answer **any two** questions from each module (2 out of 3). Each question carries 3 marks.

- 9. Explain the significance of water activity in food preservation.
- 10. Describe the forms of water in food systems.
- 11. Classify carbohydrates with suitable examples.
- 12. Explain the functional role of carbohydrates in jam preparation.
- 13. Describe the four levels of protein structure.
- 14. Explain the principle of paneer preparation by acid coagulation.
- 15. Distinguish between fats and oils.
- 16. Differentiate between fat-soluble and water-soluble vitamins.

Part C — (4 × 7 = 28 marks)

Answer **any one** question from each module (with choice). Each question carries 7 marks.

- 17. (a) Describe the determination of moisture content by oven drying method with a worked example.
  OR
  (b) Explain water activity and its role in food stability with a table showing a_w ranges and microbial growth.
- 18. (a) Discuss the chemistry, classification, and functional properties of carbohydrates in food processing.
  OR

(b) Explain the preparation of jam, including the role of sugar, pectin, and acid.

- 19. (a) Describe the structure, classification, and functional roles of proteins in food.
  OR
  (b) Explain the preparation of mayonnaise and paneer, relating protein functionality to product quality.
- 20. (a) Discuss the classification and properties of lipids with examples.
  OR
  (b) Explain the types and sources of vitamins and minerals with their functional roles.

Quick Revision

Per-module revision cards, common mistakes, and exam checklist.

Module I — Water & Nutrients

- $a_w = P/P_0$; controls microbial growth.
- Free water vs. bound water.
- Moisture (%) = $(W_i - W_d)/W_i \times 100$.
- Macros: carbs, proteins, fats (4, 4, 9 kcal/g).
- Micros: vitamins, minerals.

Module II — Carbohydrates

- General formula: $C_n(H_2O)_n$.
- Classification: mono-, di-, oligo-, polysaccharides.
- Functions: sweetening, texture, browning, preservation.
- Jam: TSS 65–68 °Brix, pH ~3.2.
- Tests: Molisch's (all carbs), Benedict's (reducing), iodine (starch).

Module III — Proteins

- Structure: Primary → Secondary → Tertiary → Quaternary.
- Classification: simple/conjugated, globular/fibrous.
- Functions: emulsification, gelation, foaming, water holding.
- Protein (%) = N (%) × 6.25.
- Paneer: acid coagulation at pH 4.6.

Module IV — Lipids, Vitamins, Minerals

- Lipids: 9 kcal/g; fats (solid) vs oils (liquid).
- Vitamins: Fat-soluble (A, D, E, K) & Water-soluble (B, C).
- Minerals: Macro (Ca, P, Mg, Na, K) & Trace (Fe, Zn, I, Se, Cu).
- Toffee: sugar + fat → smooth texture, rich mouthfeel.
- Sauerkraut: lactic acid fermentation.

Common Mistakes to Avoid

- **Confusing a_w with moisture content:** a_w measures free water availability, not total water present.
- **Mixing up reducing and non-reducing sugars:** Sucrose is non-reducing (no free aldehyde/ketone); glucose and fructose are reducing.
- **Forgetting the conversion factor:** Protein (%) = Nitrogen (%) × 6.25 — not 6.25% directly.
- **Confusing fat-soluble vs water-soluble vitamins:** A, D, E, K are fat-soluble; B-complex and C are water-soluble.
- **Oven drying temperature:** Use 105°C, not 100°C or higher than 110°C.

Exam Checklist

- ✅ Know the definitions of a_w, moisture content, TSS.
- ✅ Memorise the energy values: 4, 4, 9 kcal/g.
- ✅ Be able to classify carbohydrates, proteins, lipids, vitamins, minerals.
- ✅ Understand the functional roles of each nutrient in food processing.
- ✅ Know the practical procedures: jam, paneer, mayonnaise, toffee, sauerkraut.
- ✅ Practice numericals: moisture content, protein estimation, TSS.
- ✅ Review the CO–PO mapping and assessment scheme.

Glossary of Important Terms

Term	Meaning
Water activity (a_w)	Ratio of partial pressure of water in food to pure water at same temperature; measures available water for microbial growth.
Macronutrient	Nutrient required in large amounts (g/day): carbohydrates, proteins, lipids.
Micronutrient	Nutrient required in small amounts (mg/µg/day): vitamins, minerals.
Monosaccharide	Simplest carbohydrate, cannot be hydrolysed further (e.g., glucose).
Polysaccharide	Complex carbohydrate composed of many monosaccharide units (e.g., starch).
TSS (Total Soluble Solids)	Measure of dissolved solids (mainly sugars) in solution, expressed in °Brix.
Denaturation	Loss of protein's secondary, tertiary, or quaternary structure without breaking peptide bonds.
Coagulation	Irreversible aggregation of denatured proteins (e.g., paneer).
Emulsification	Process of stabilising an oil-in-water or water-in-oil mixture using an emulsifier (e.g., protein, lecithin).
Fermentation	Microbial conversion of sugars to acids, alcohols, or gases (e.g., sauerkraut, yogurt).
Rancidity	Oxidative or hydrolytic breakdown of lipids leading to off-flavours.
Maillard reaction	Non-enzymatic browning between reducing sugars and amino acids.

References and Further Reading

Textbooks, references, and web resources listed in the syllabus.

- **Textbooks:**
  - — Fennema, O.R. (2007). *Food Chemistry*. 4th ed. CRC Press.
  - — Potter, N.N. & Hotchkiss, J.H. (1995). *Food Science*. 5th ed. Springer.
  - — Manay, S. & Shadaksharaswamy, M. (2008). *Foods: Facts and Principles*. New Age International.
- **Reference books:**
  - — Belitz, H.D., Grosch, W. & Schieberle, P. (2009). *Food Chemistry*. 4th ed. Springer.
  - — Fellows, P.J. (2009). *Food Processing Technology*. 3rd ed. Woodhead Publishing.
- **Web resources (as listed in syllabus):**
  - — NPTEL courses on Food Processing and Technology
  - — FAO (Food and Agriculture Organization) publications on food science